

Online-offline Top Quark Tagging

Smearing Effects and Resolution Recovery

Tianshu Gu
2026/5/5

Physics Background and Goal of this work

The **top quark** is the heaviest particle in the Standard Model, which plays a key role in many searches for new physics.

Online HLT jets

- real-time
- Low resolution
 - limited detector resolution
 - real-time reconstruction constraints
 - simplified algorithms

Offline reconstructed jets

- full detector information
- high resolution

Final goal: Recover **offline-quality jet constituent information** from degraded **HLT-level jets** to improve downstream **top tagging performance**.

Contents

- Big Picture: Full HLT Simulation Pipeline
- Focus: Smearing Effects and Resolution Recovery
- Models, Results and Analysis
- Additional Experiments

Big Picture: Full HLT Simulation Pipeline

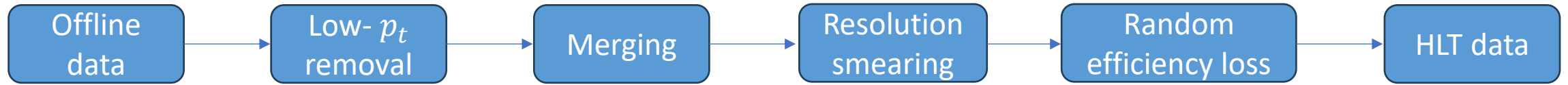
Dataset and HLT Simulation

Offline jets: <https://opendata.cern.ch/record/15013> (J,C,D)

Number of jets Number of constituents per jet

Per-constituent features
Mainly use these four-dimensional features:
 p_T, η, ϕ, E

HLT jets: simulated from offline jets (All HLT effects are applied at the constituent level.)



- Low- p_T removal: Remove constituents with $p_T < p_{T_threshold}(HLT)$
- Merging: Nearby particles are merged if $\Delta R < R_{merge}$, $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$. Only keep the merged constituent.

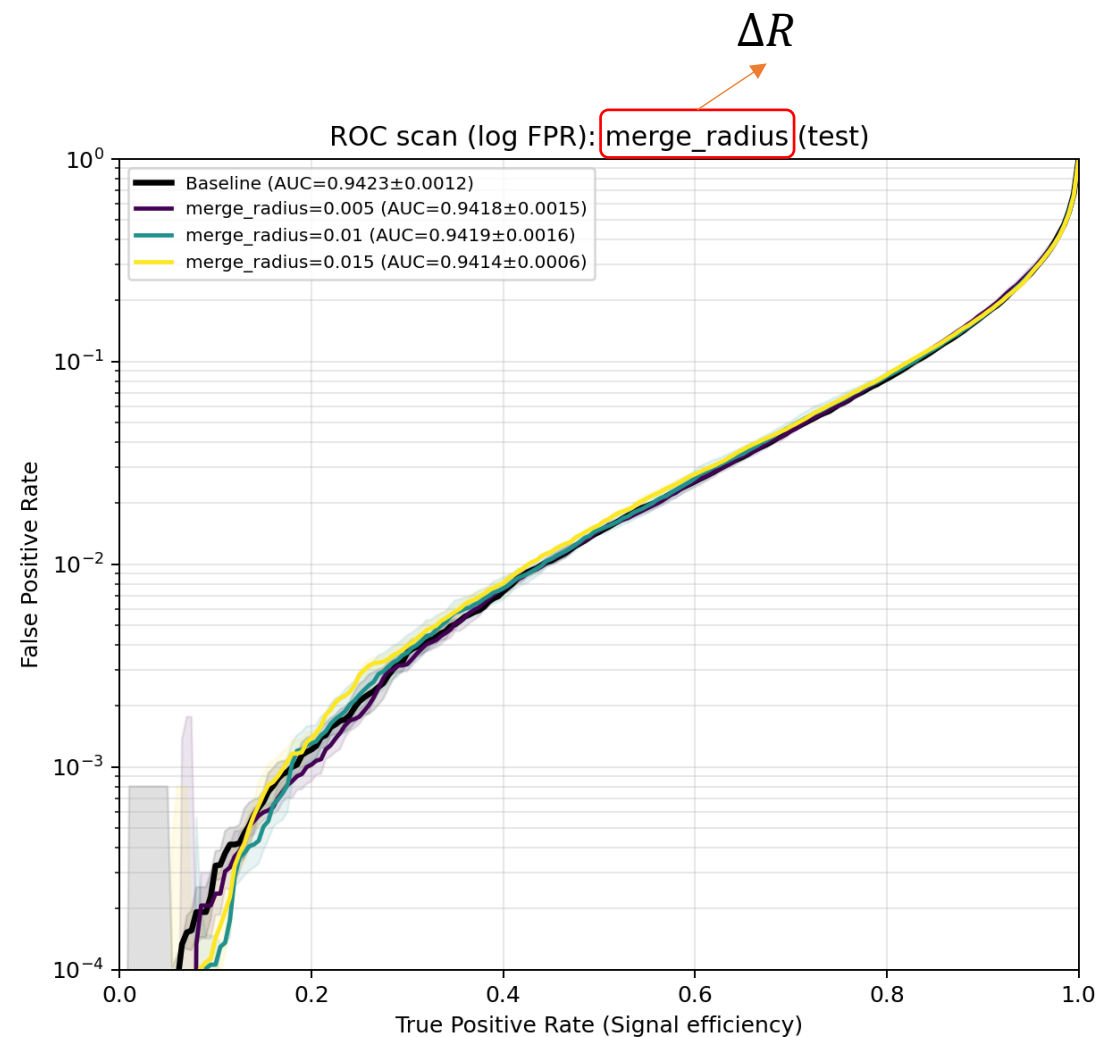
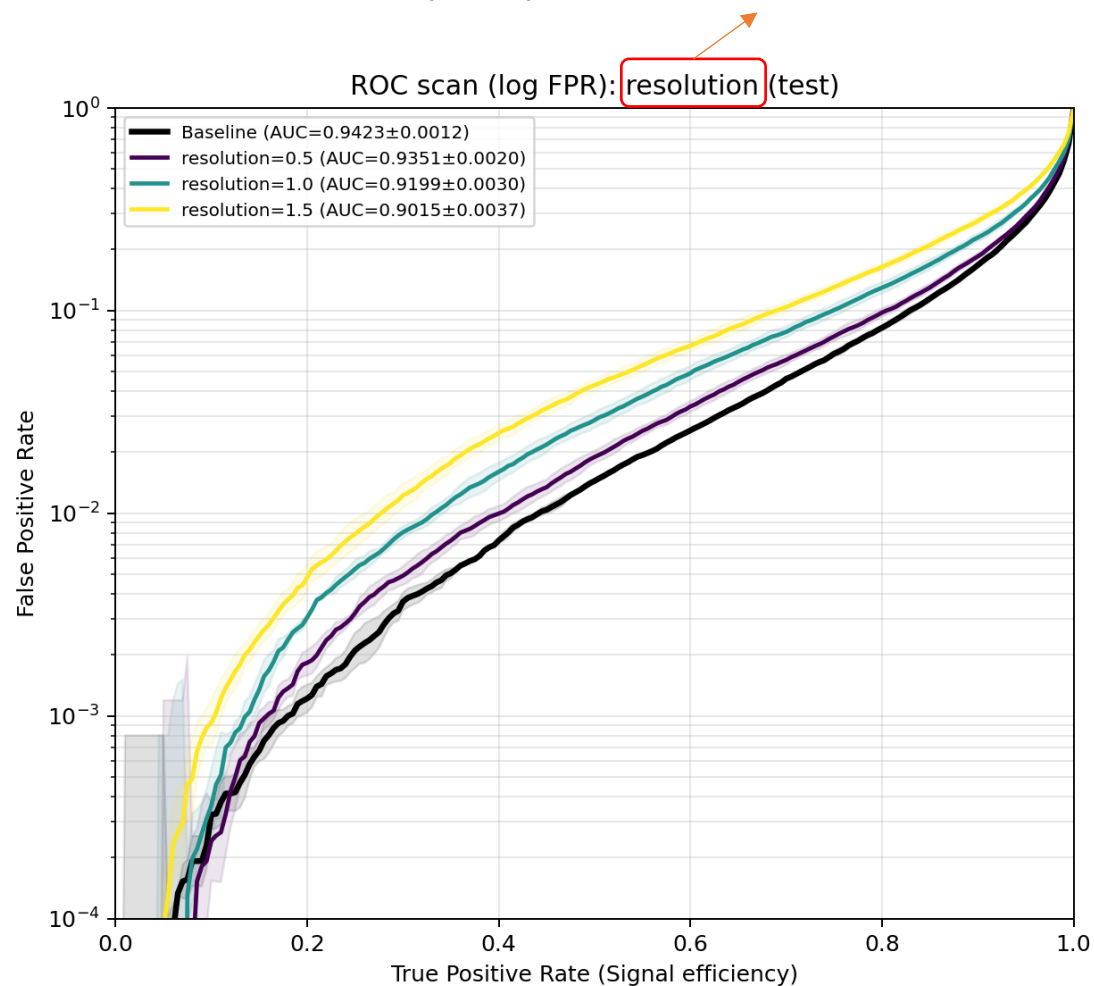
Merged constituent properties would be:

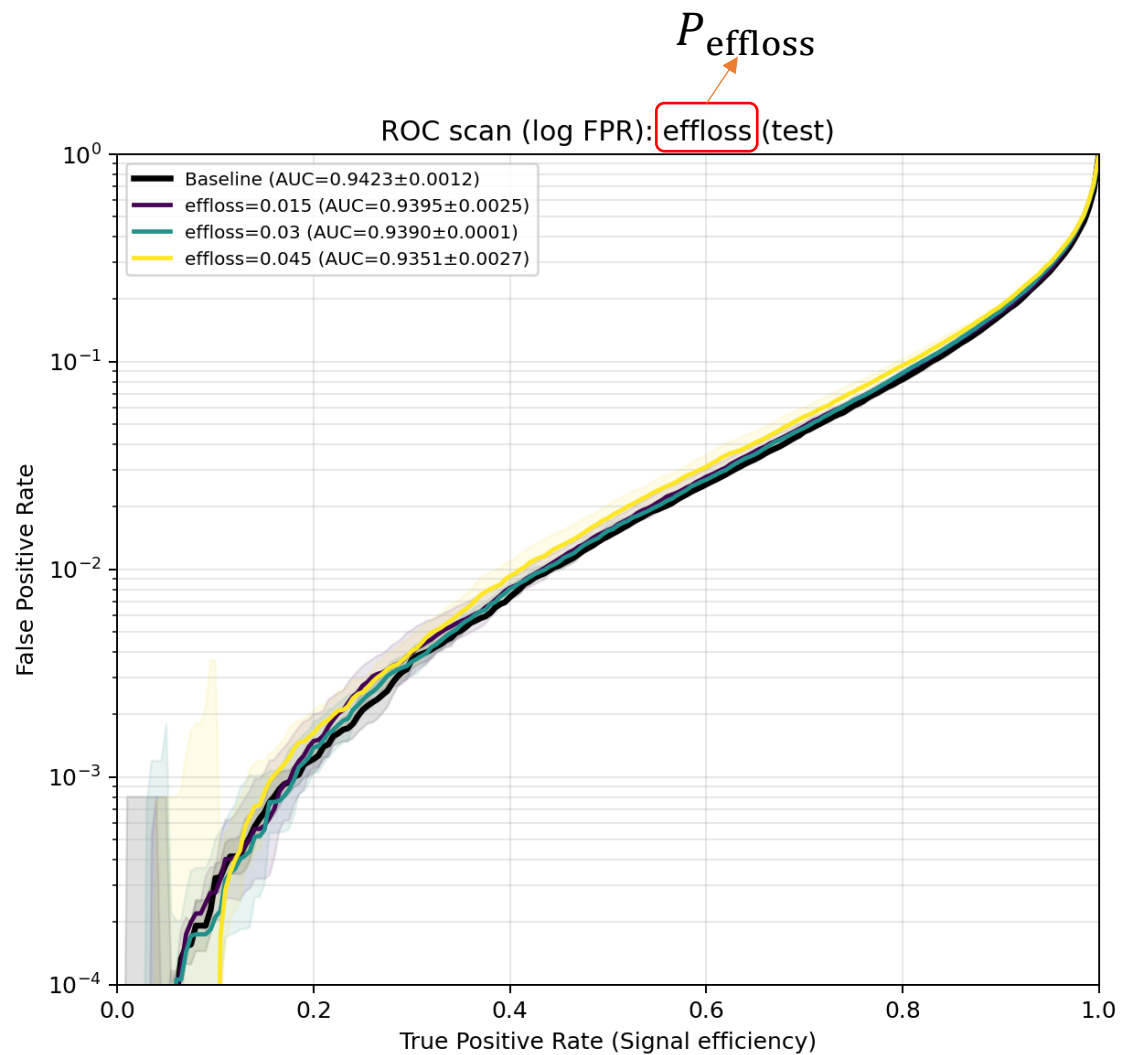
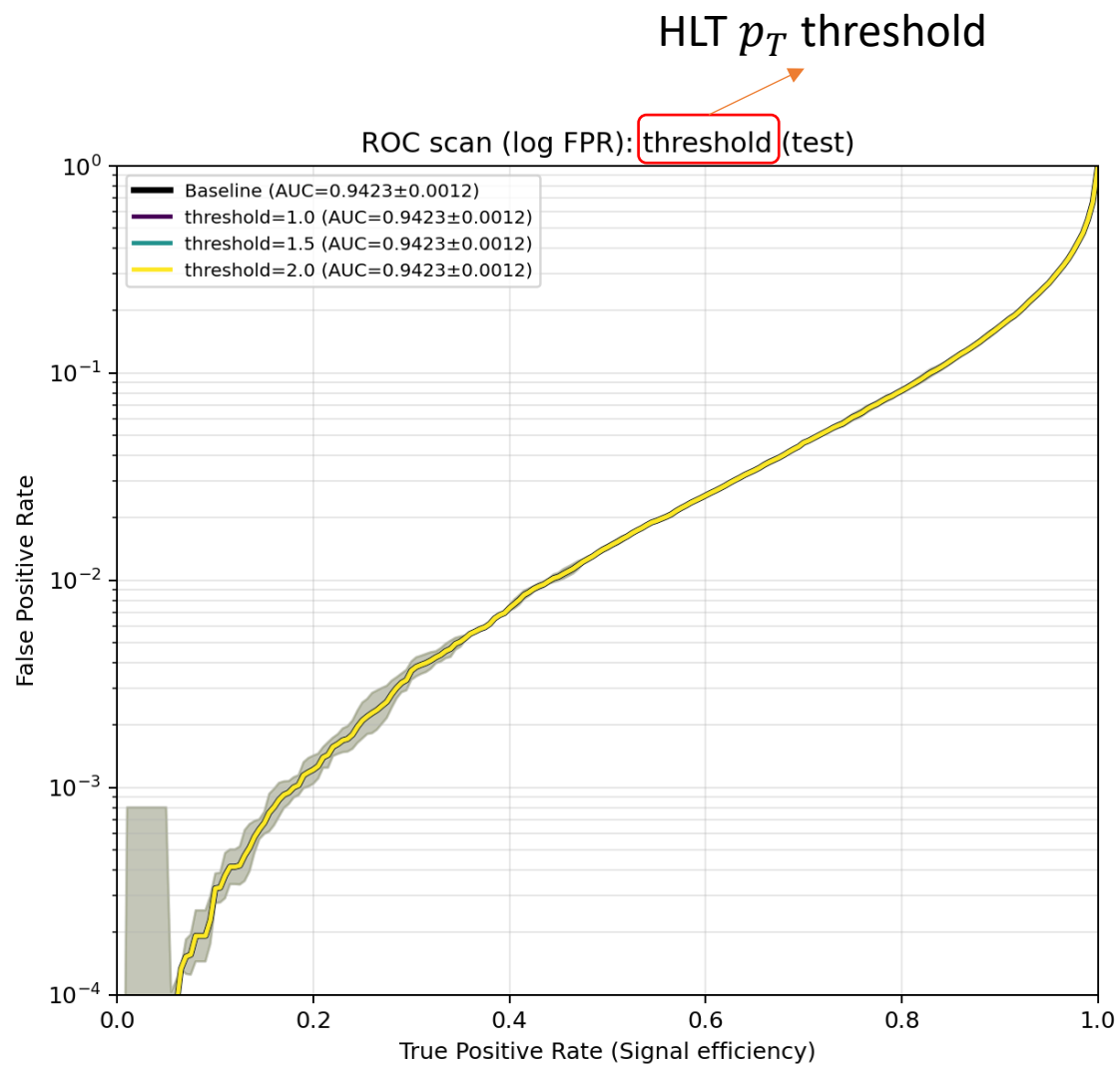
- $p_T = p_{T1} + p_{T2}$
- $\eta = w_1\eta_1 + w_2\eta_2$ ($w_{1,2} = \frac{p_{T1,2}}{p_{T1}+p_{T2}}$)
- $\phi = \text{atan2}(w_1\sin\phi_1 + w_2\sin\phi_2, w_1\cos\phi_1 + w_2\cos\phi_2)$ ($w_{1,2} = \frac{p_{T1,2}}{p_{T1}+p_{T2}}$)
- $E = E_1 + E_2$
- Resolution smearing:
 - $p'_T = p_T \times \mathcal{N}(1, \sigma_{p_T})$
 - $\eta' = \eta + \mathcal{N}(0, \sigma_\eta)$
 - $\phi' = \phi + \mathcal{N}(0, \sigma_\phi)$
 - $E' = p_T \cosh(\eta)$ → E' of HLT constituent only depends on p_T and η .
- Random efficiency loss: randomly remove constituents with probability $P_{effloss}$

Controlled Study of HLT Effects

Baseline configuration: constituents with $p_T < 0.5$ are removed for both offline and HLT data. No other HLT effects.

Resolution acts as a scale factor: base values $\sigma_{p_T} = 0.1$ and $\sigma_\eta = \sigma_\phi = 0.03$, scaled by $[0.5, 1.0, 1.5]$.

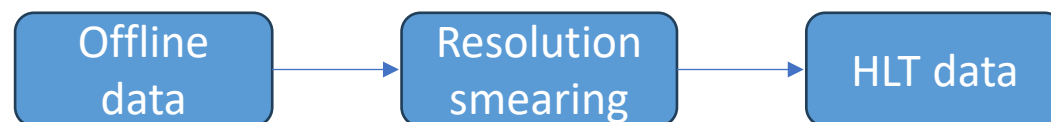




Conclusion: resolution smearing appears to be the primary source of information loss and AUC degradation.

Focus: Smearing Effects and Resolution Recovery

Smearing Effects and Resolution Recovery



Motivation

- HLT processing introduces resolution degradation and information loss.
- Recovering smeared jet features may improve jet representations.
- Better representations could improve downstream tagging performance.

Preprocessing

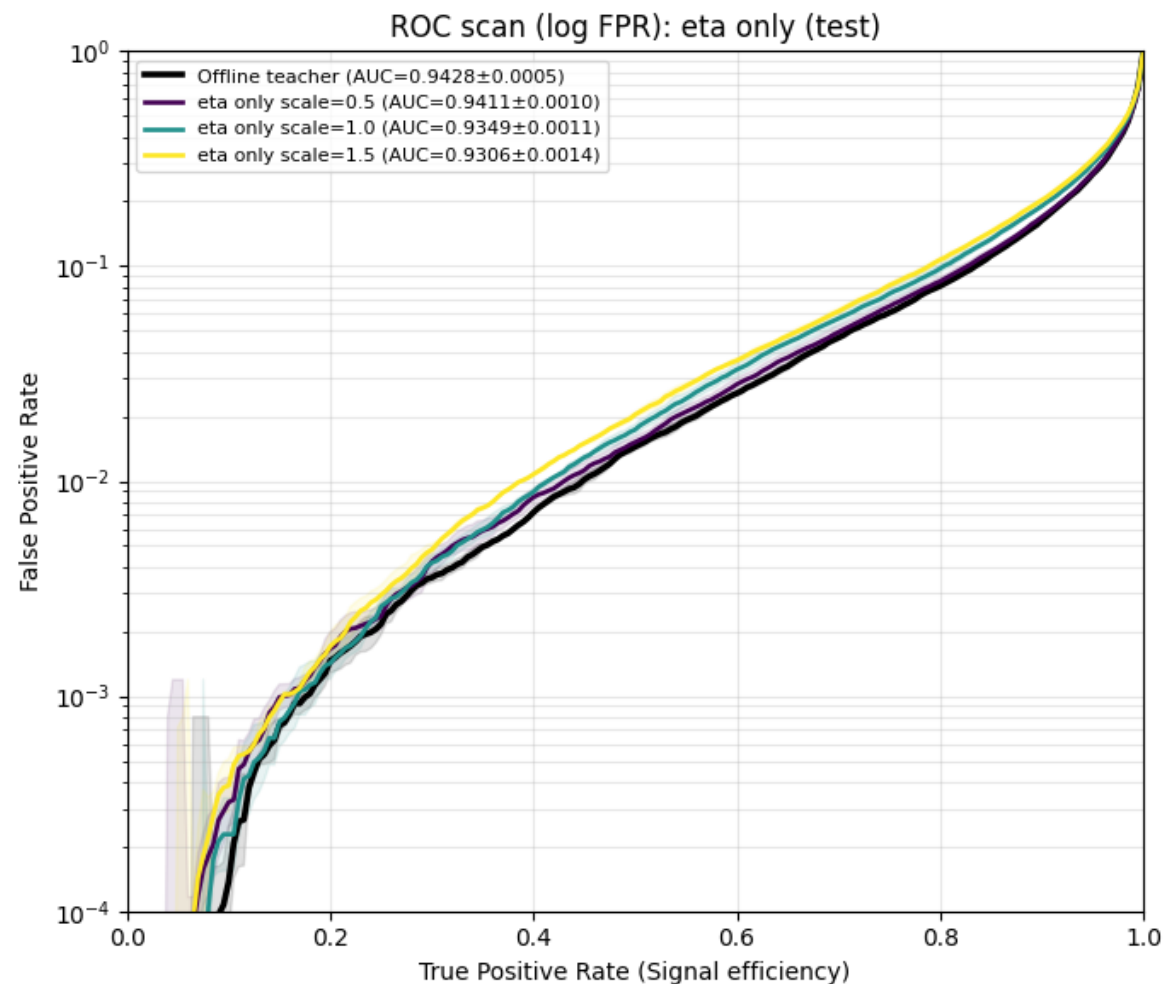
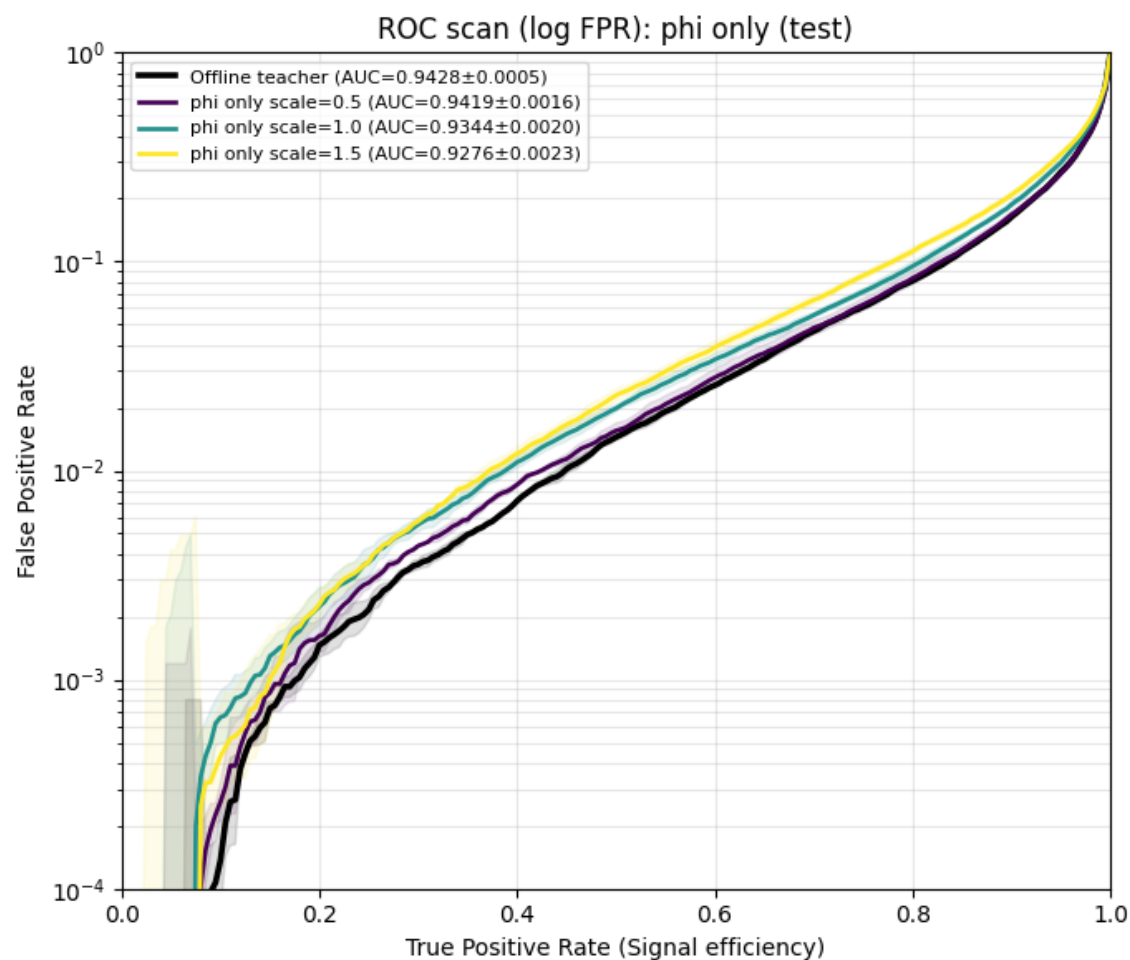
Data Preprocessing (Input should be four-dimensional features $[p_T, \eta, \phi, E]$.)

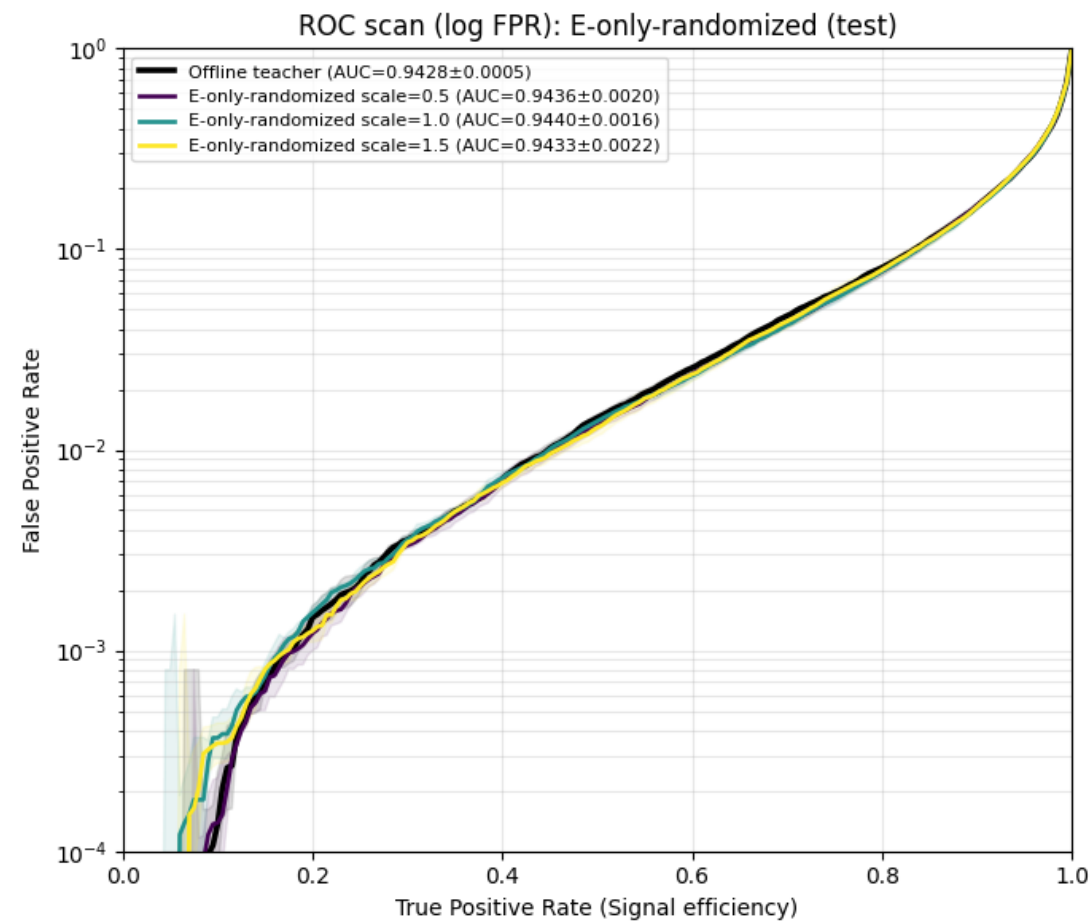
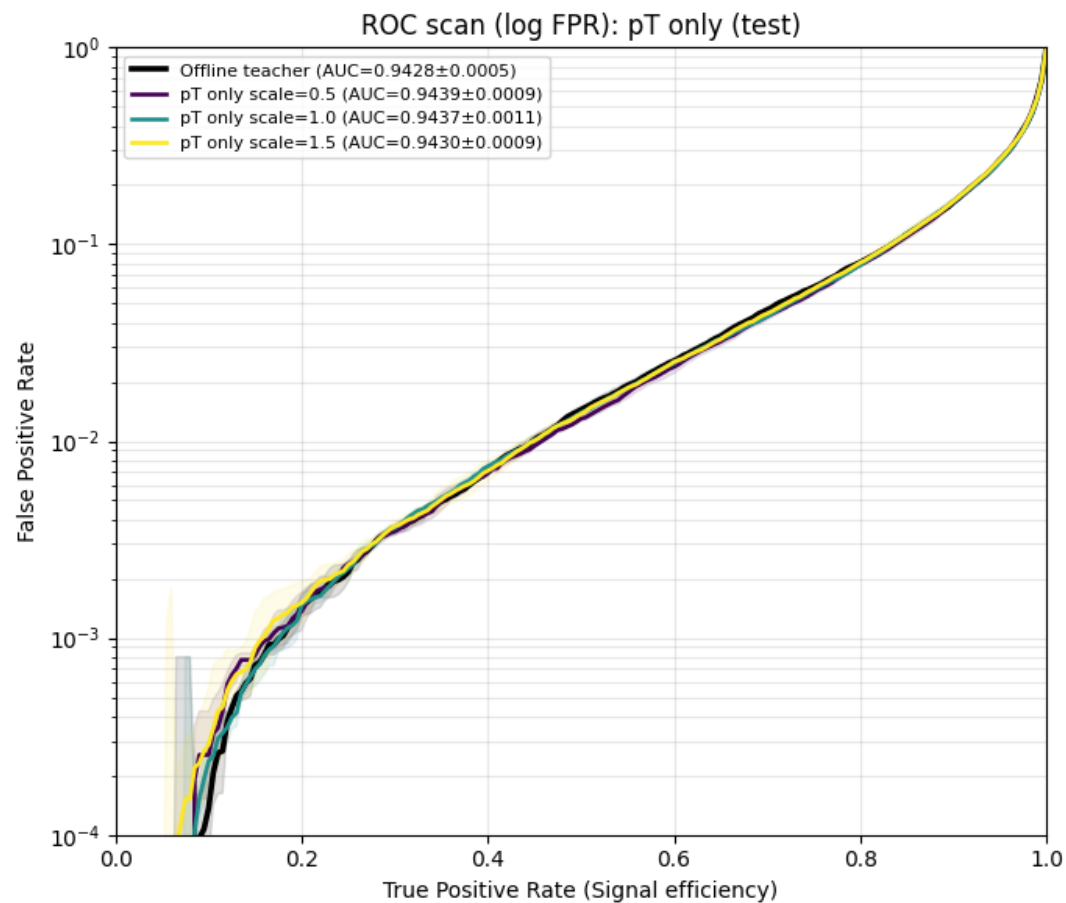
- Transform constituent features into the jet-axis frame and construct a stable relative feature representation. (For simplicity, physical quantities at the jet level are denoted with a subscript j .)
 - $p_x, p_y, p_z = p_T \cos \phi, p_T \sin \phi, p_T \sinh \eta$
 - $p_{jx}, p_{jy}, p_{jz} = \sum(p_{x,y,z} * \text{mask}), E_j = \sum(E * \text{mask})$
 - $p_{jT} = \sqrt{p_{jx}^2 + p_{jy}^2}, p_j = \sqrt{p_{jx}^2 + p_{jy}^2 + p_{jz}^2}$
 - $\eta_j = \frac{1}{2} \ln \left(\frac{p_j + p_{jz}}{p_j - p_{jz}} \right), \phi_j = \text{atan2}(p_{jy}, p_{jx})$
 - $\Delta\eta = \eta - \eta_j, \Delta\phi = \text{atan2}(\sin(\phi - \phi_j), \cos(\phi - \phi_j))$
 - Output: $[\Delta\eta, \Delta\phi, \ln(p_T), \ln(E), \ln \left(\frac{p_T}{p_{jT}} \right), \ln \left(\frac{E}{E_j} \right), \sqrt{\Delta\eta^2 + \Delta\phi^2}]$
- Standardization

$$\text{feature}' = \frac{\text{feautre} - \mu}{\sigma}$$

Controlled Study of Smearing Effects

Scale: base values $\sigma_{p_T} = 0.1$ and $\sigma_\eta = \sigma_\phi = 0.03$, scaled by [0.5,1.0,1.5].



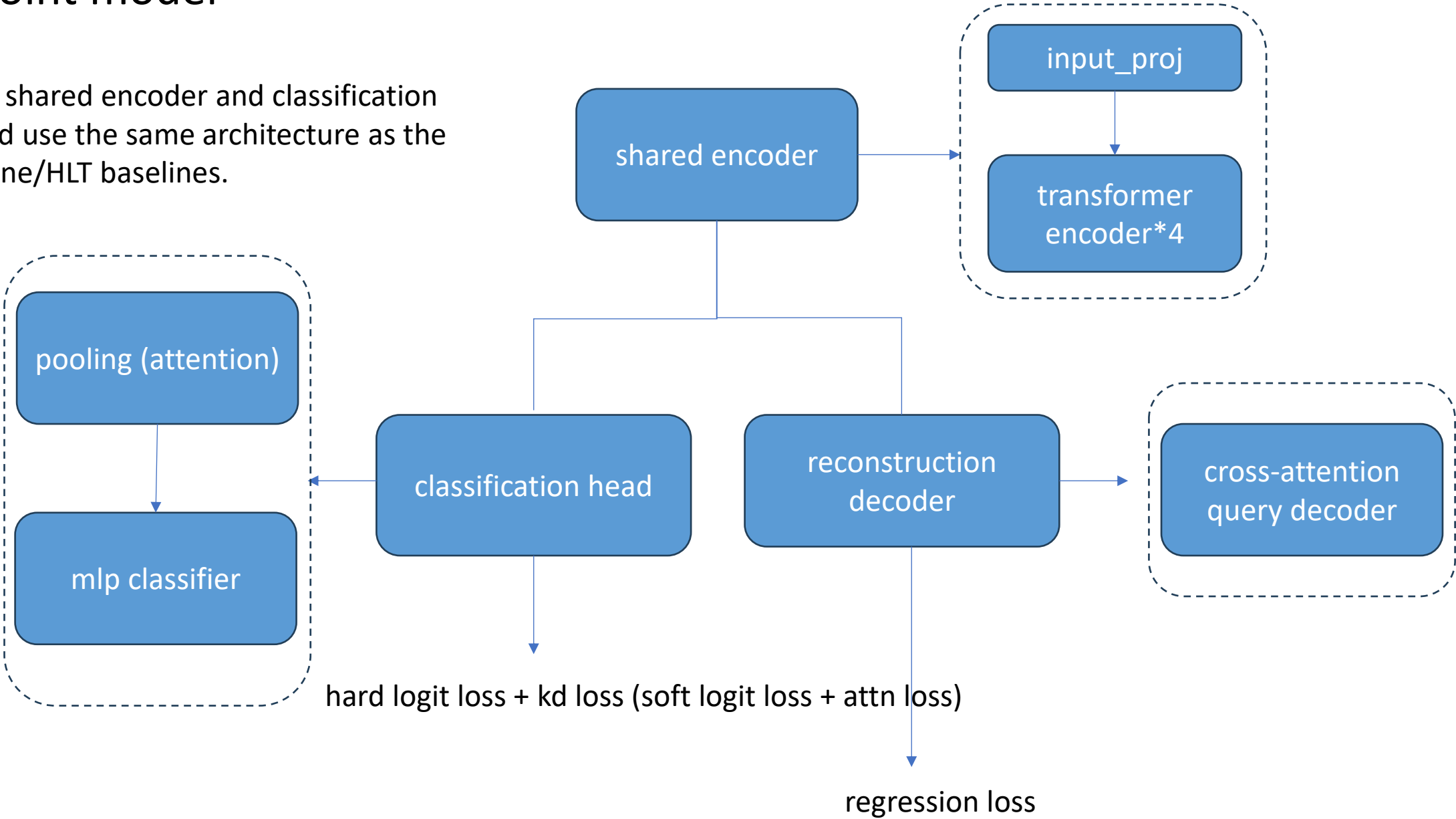


It appears that the smearing in the angular features is the primary driver of the degradation in classification AUC.

Models, Results and Analysis

Joint model

The shared encoder and classification head use the same architecture as the offline/HLT baselines.



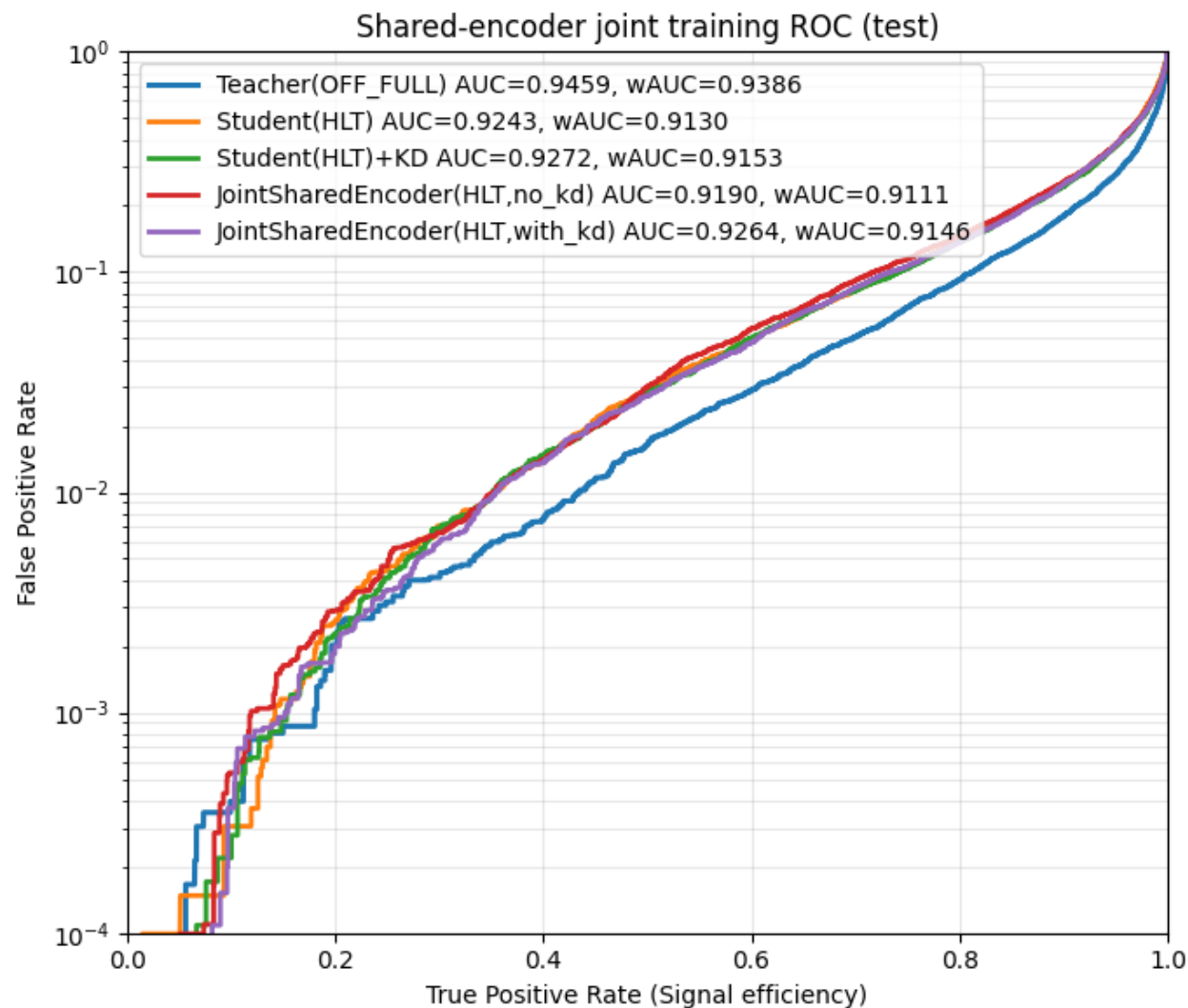
Motivation

- Avoid error propagation in sequential reconstruction-classification pipelines.
- Joint training allows reconstruction and classification objectives to guide the encoder simultaneously.
- Shared representations may improve both feature recovery and tagging performance.

Training Setup

- Smearing is re-sampled every epoch
- training converges with early stopping
- dataset splits are identical across runs.

200k (0.7/0.15/0.15)

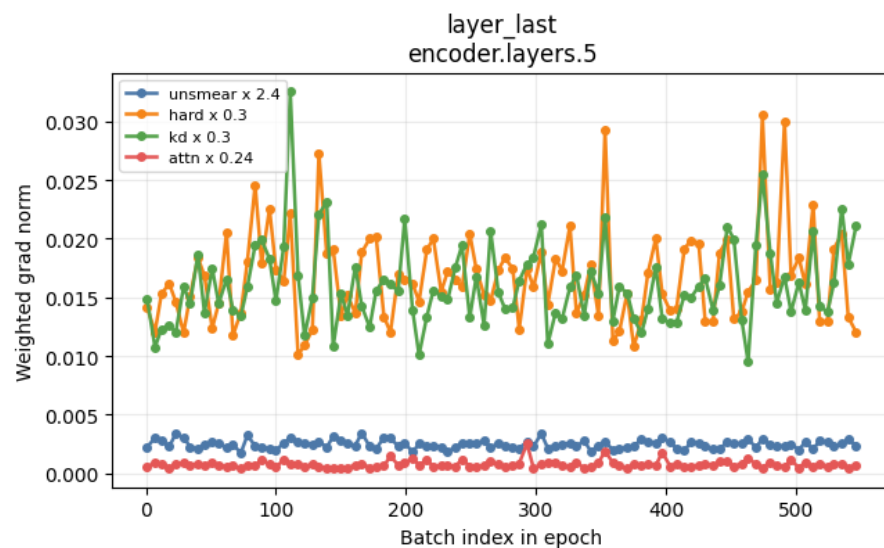
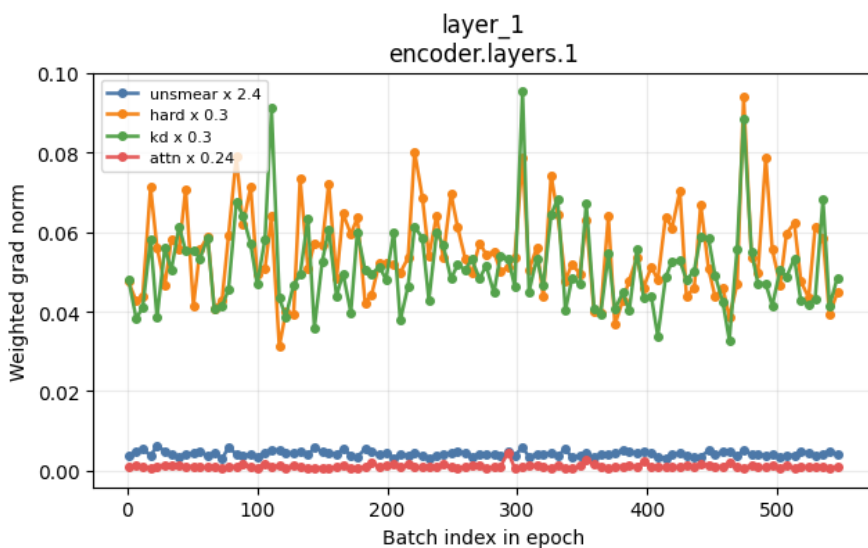
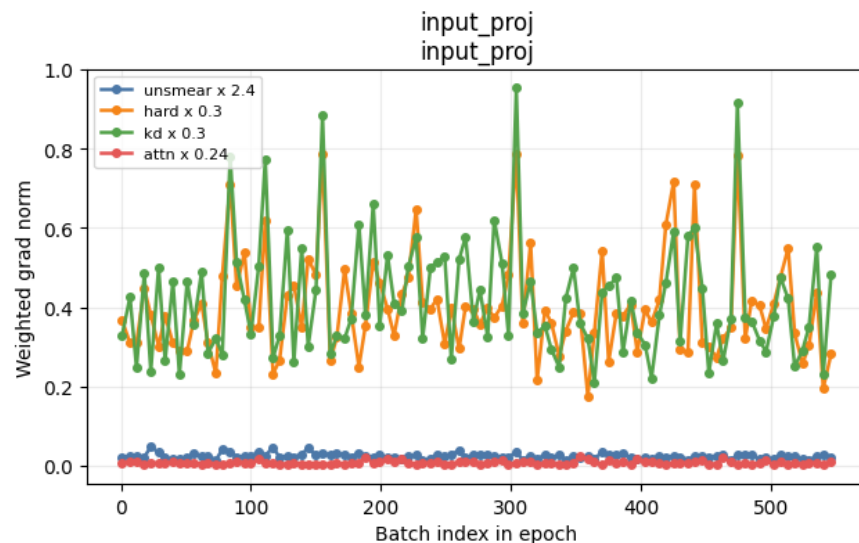
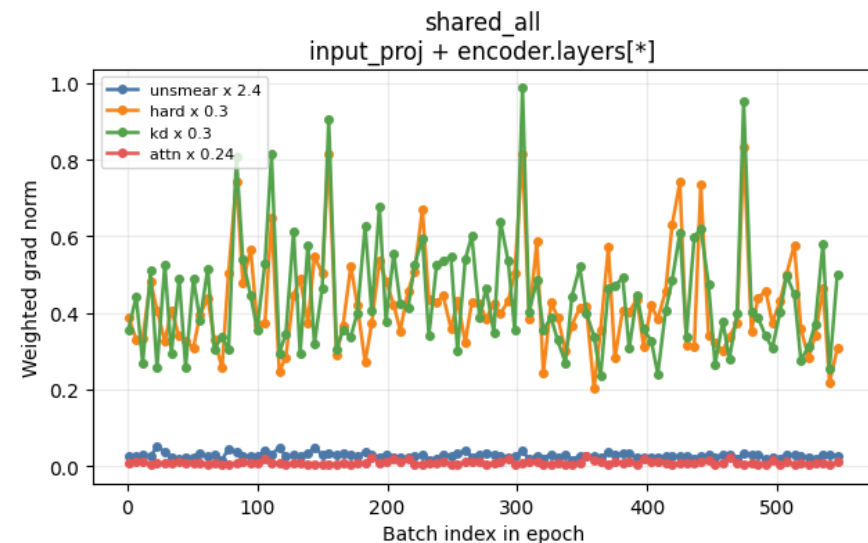


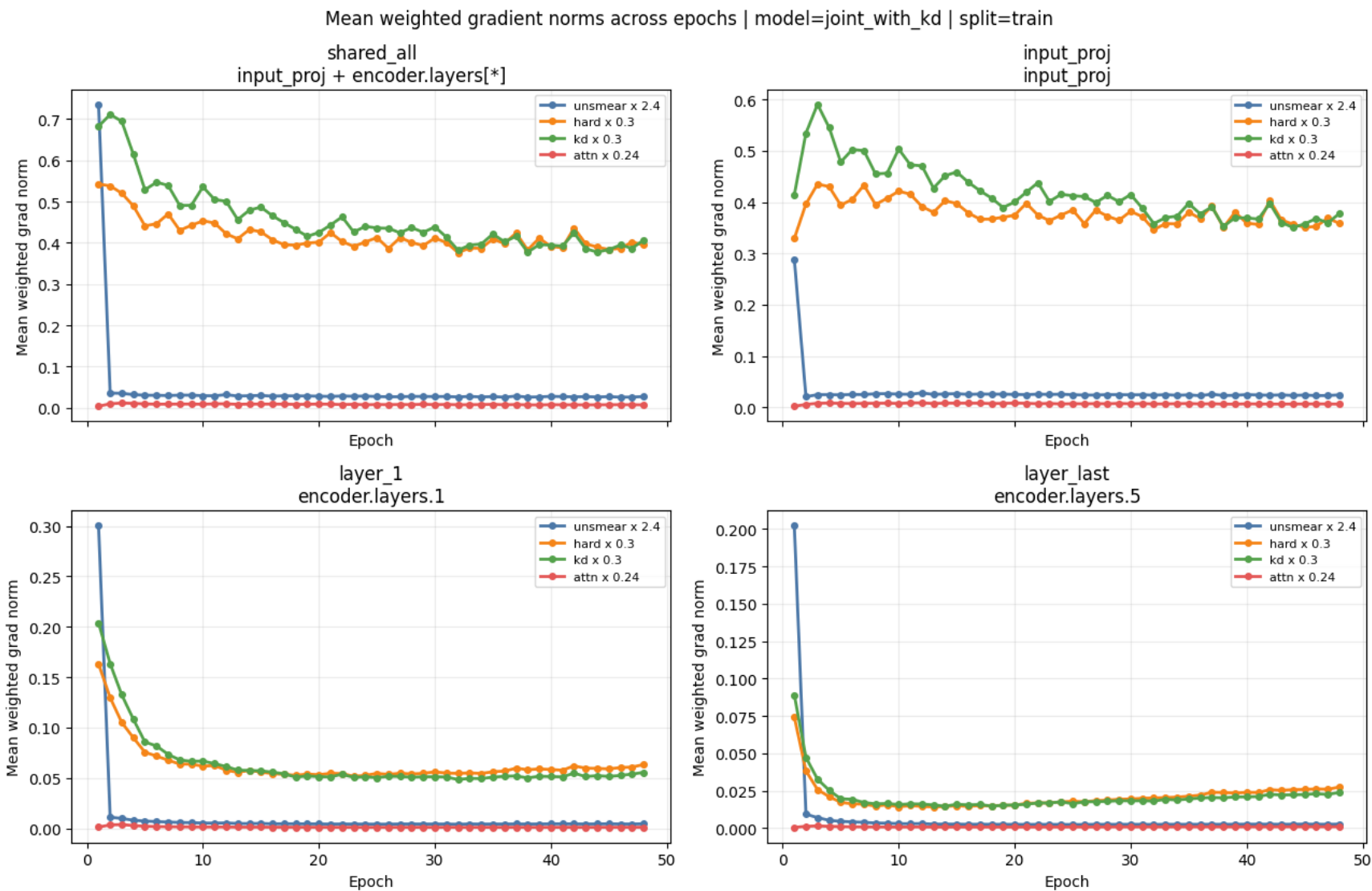
Under this design, the reconstruction branch does not lead to improved classification AUC.

Gradient analysis

“Unsmear” = reconstruction task in the following slides.

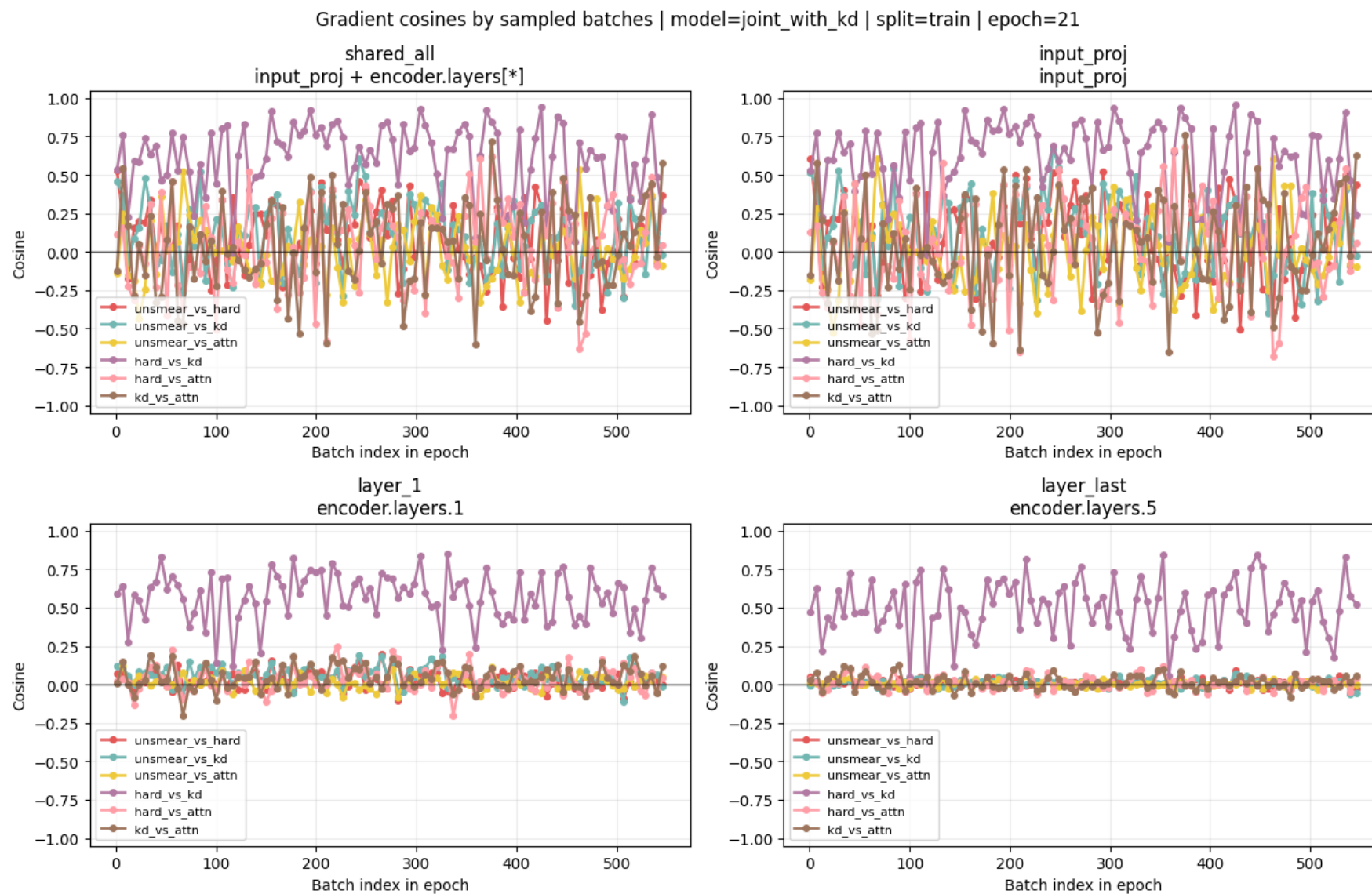
Weighted gradient norms by sampled batches | model=joint_with_kd | split=train | epoch=21



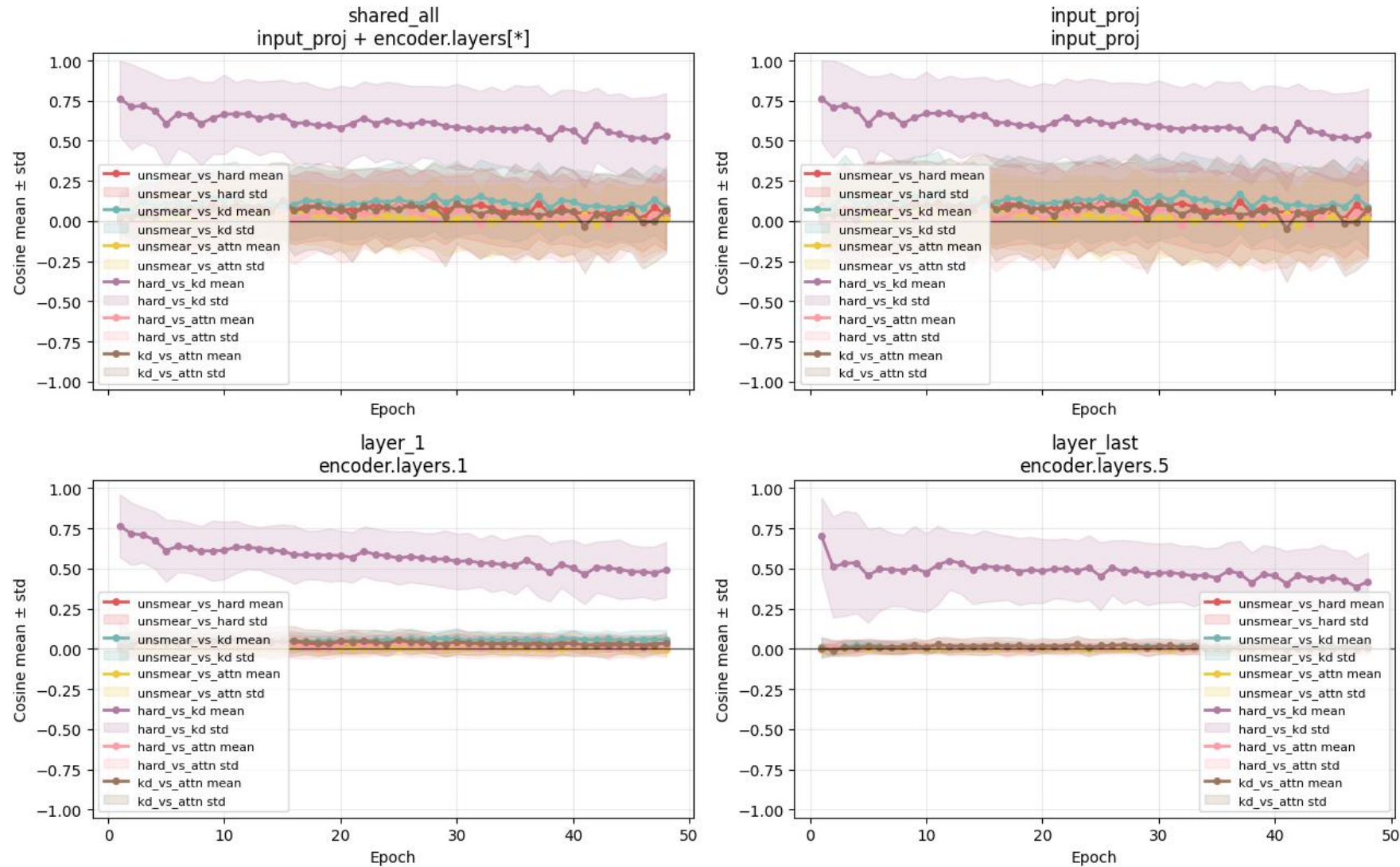


The reconstruction branch has limited influence on the shared encoder. Increasing the reconstruction loss weight mainly affects the early training stage, while its impact becomes negligible in later epochs.

Gradient cosine similarity between reconstruction and classification gradients



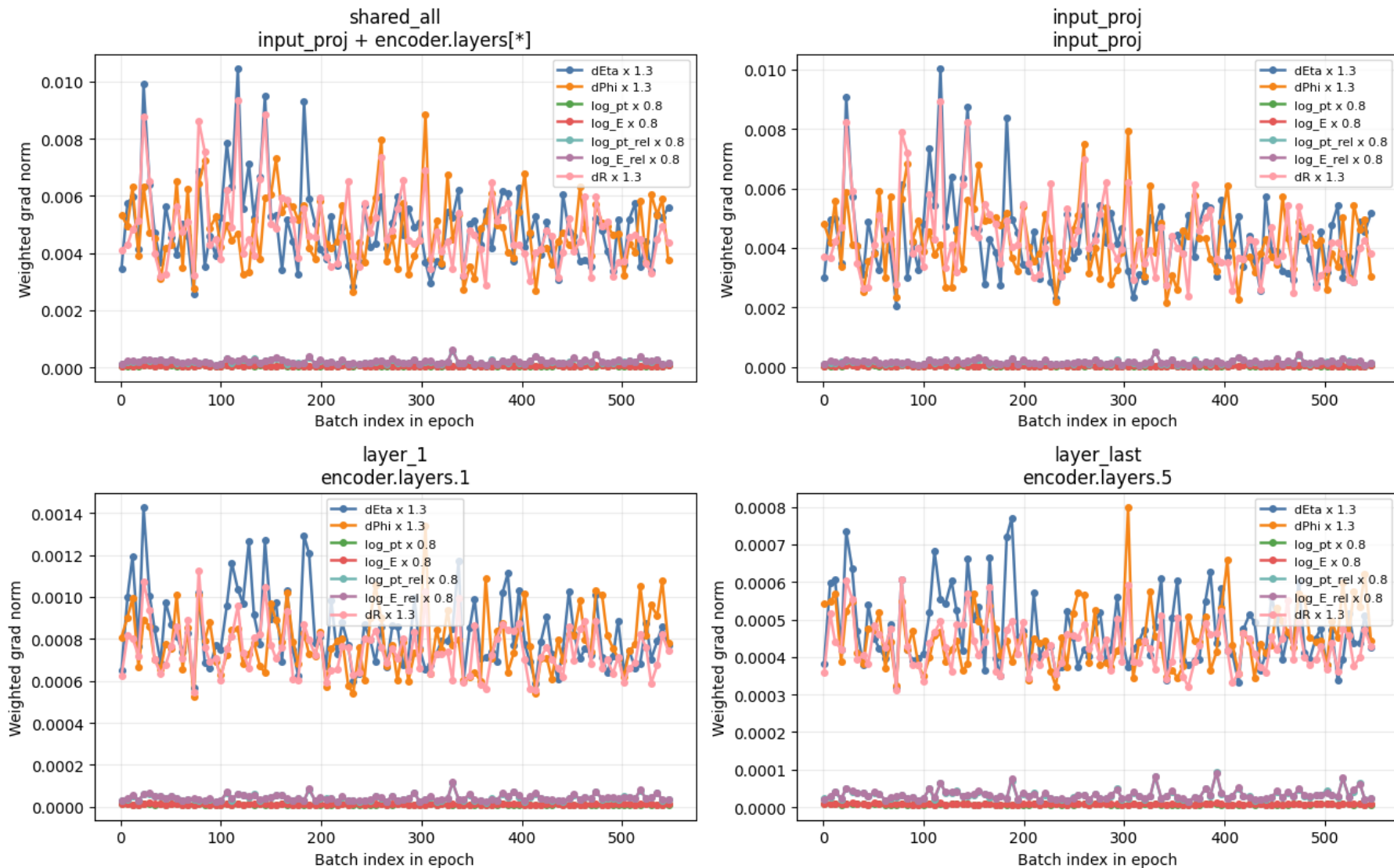
Mean and std of gradient cosines across epochs | model=joint_with_kd | split=train



- Positive: aligned objectives
- Negative: conflicting optimization directions
- Near zero: weak interaction between branches

The gradient cosine remains close to zero overall, suggesting that the reconstruction branch has limited influence on the shared encoder optimization.

Weighted feature gradient norms by sampled batches | model=joint_with_kd | split=train | epoch=21

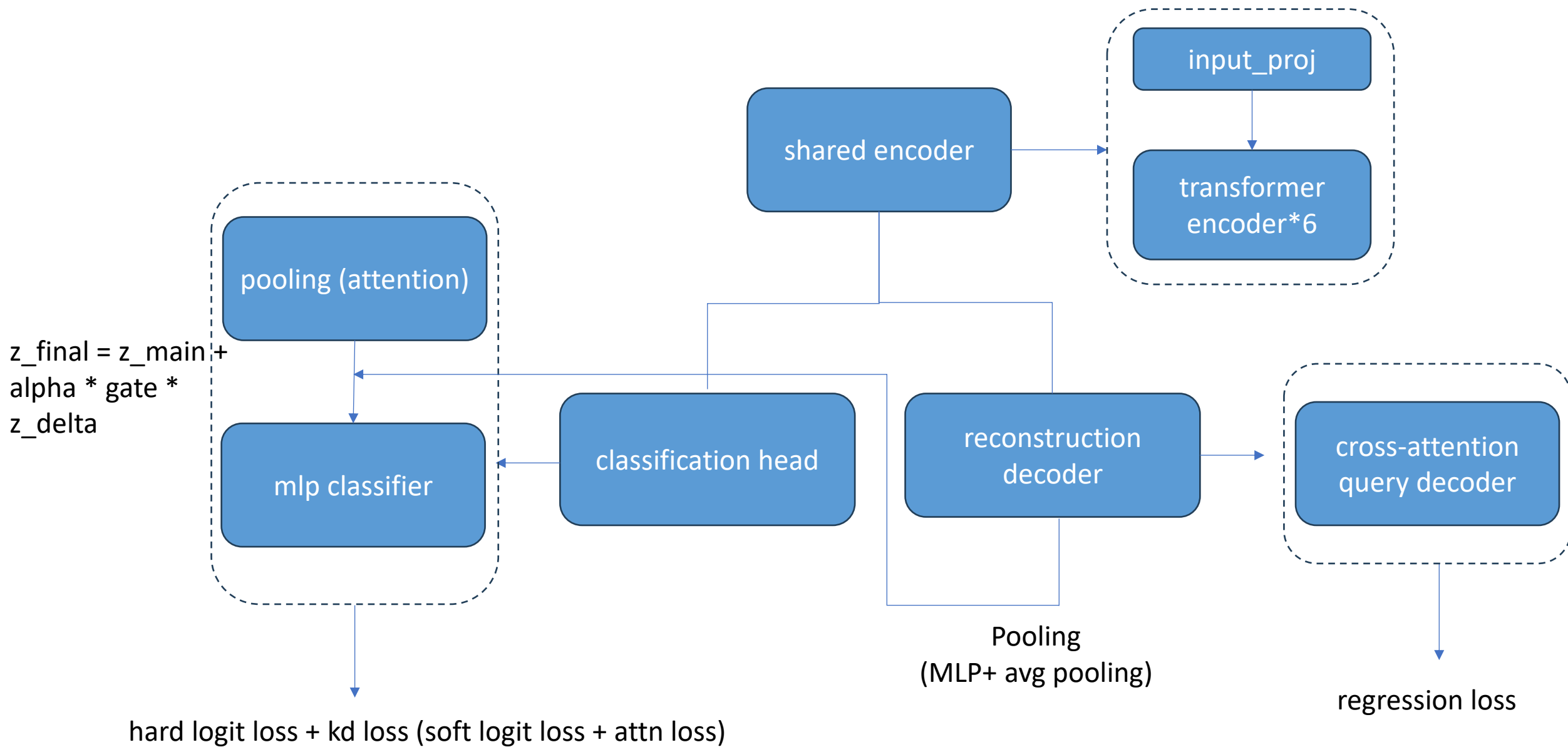


Angular losses show the largest influence, but their effect remains limited because the reconstruction objective itself contributes very weakly to the shared encoder optimization.

A key issue of the standard joint model is that the reconstruction objective has only a very limited influence on the classification task. (Since the reconstruction branch influences classification only through the shared encoder, its weak impact on the shared encoder results in limited downstream effect on classification.)

Another key issue is that the reconstruction objective even leads to worse downstream classification performance.

Joint model with fusion



Motivation

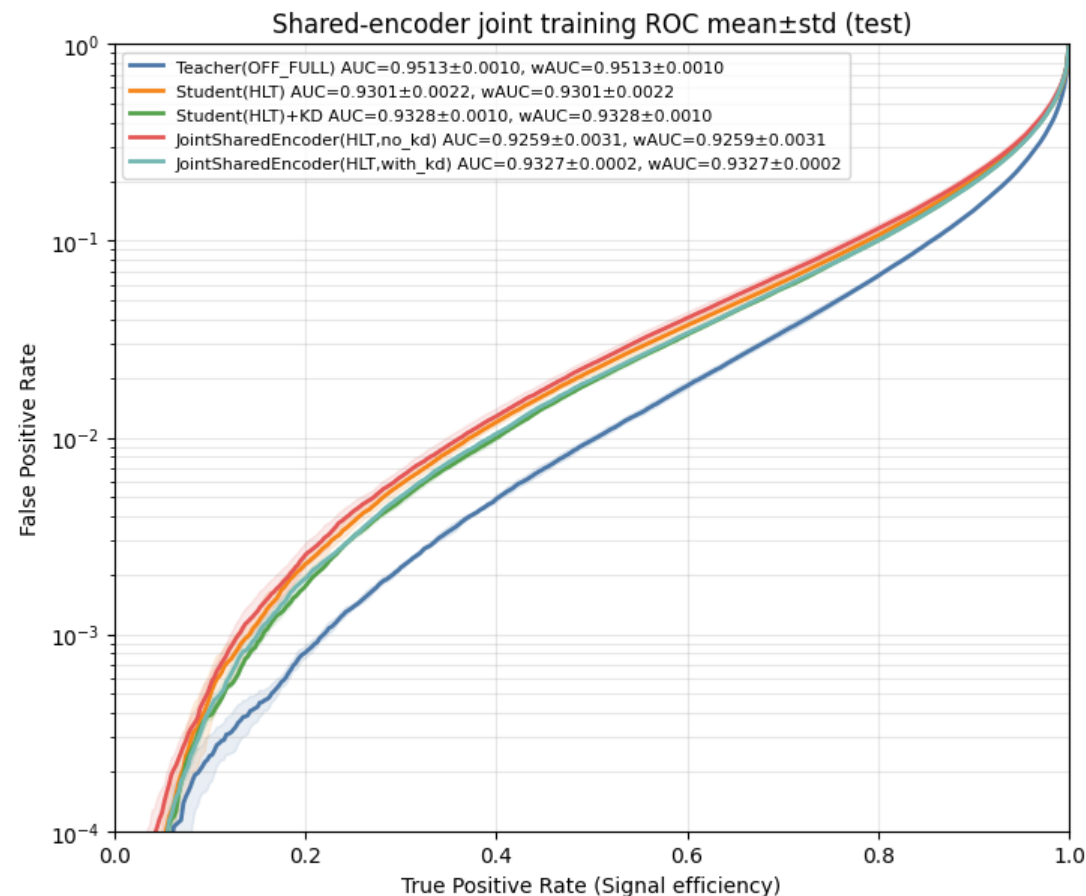
- Increase the influence of the reconstruction task on downstream classification through feature fusion.
- Use gated fusion to incorporate reconstruction information only when beneficial, aiming to avoid degradation and maintain performance $\geq$ HLT.

Same training setup.

1.5M (1M/0.25M/0.25M)

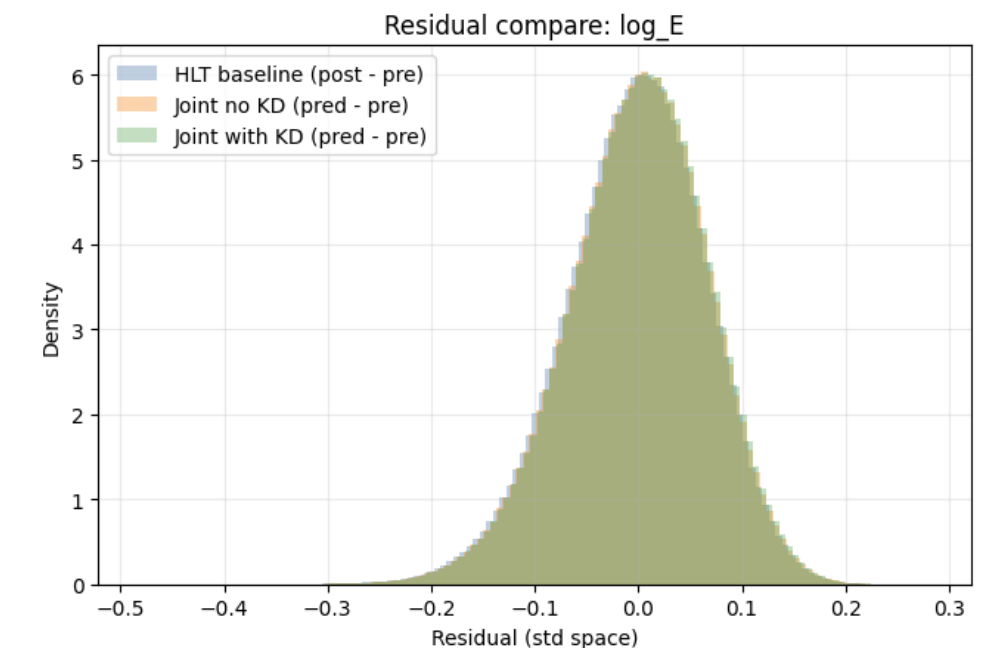
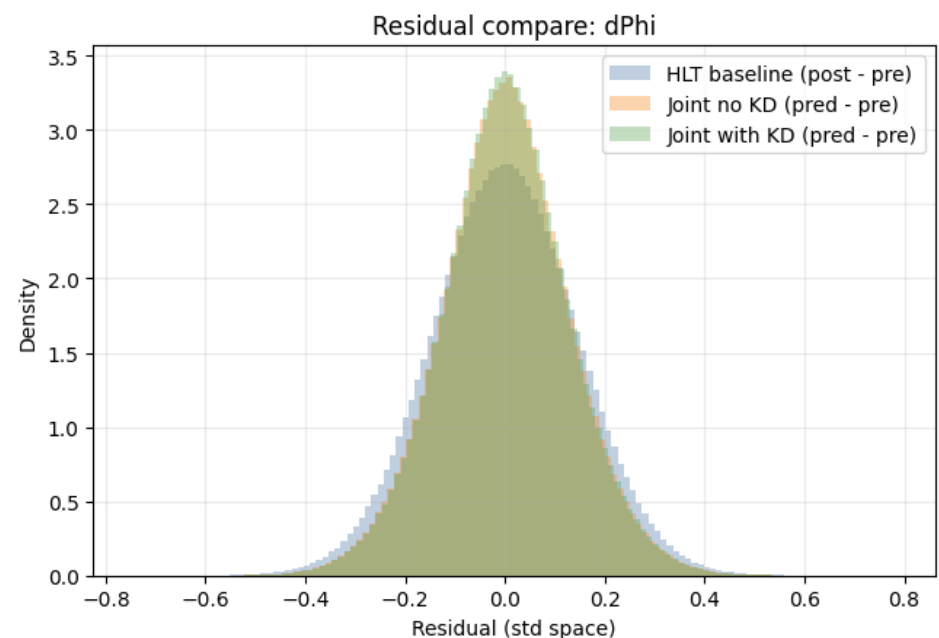
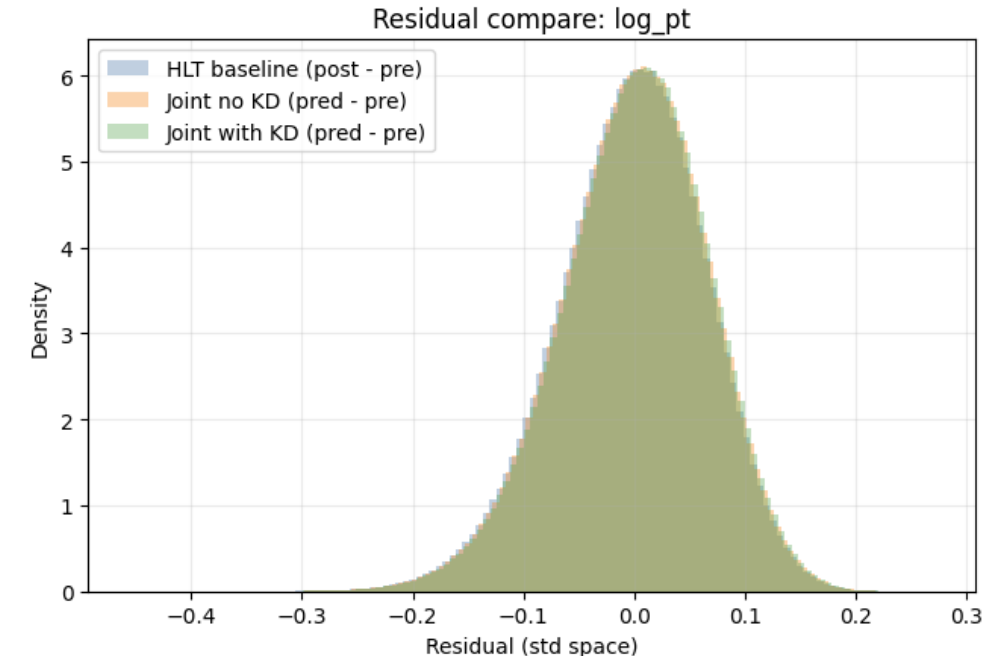
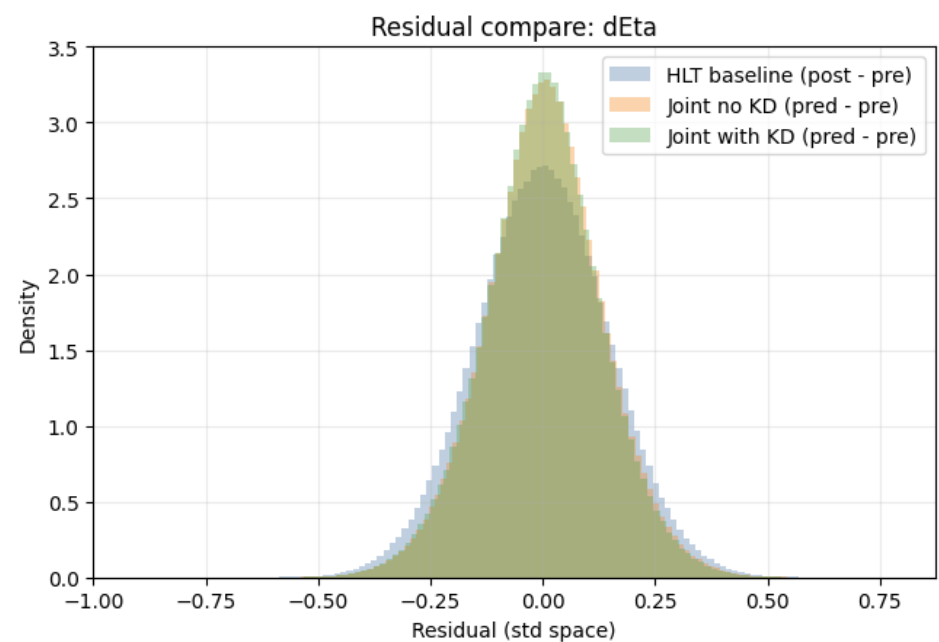
Increase the training dataset size from 200k to 1.5M to reduce statistical fluctuations and improve training stability.

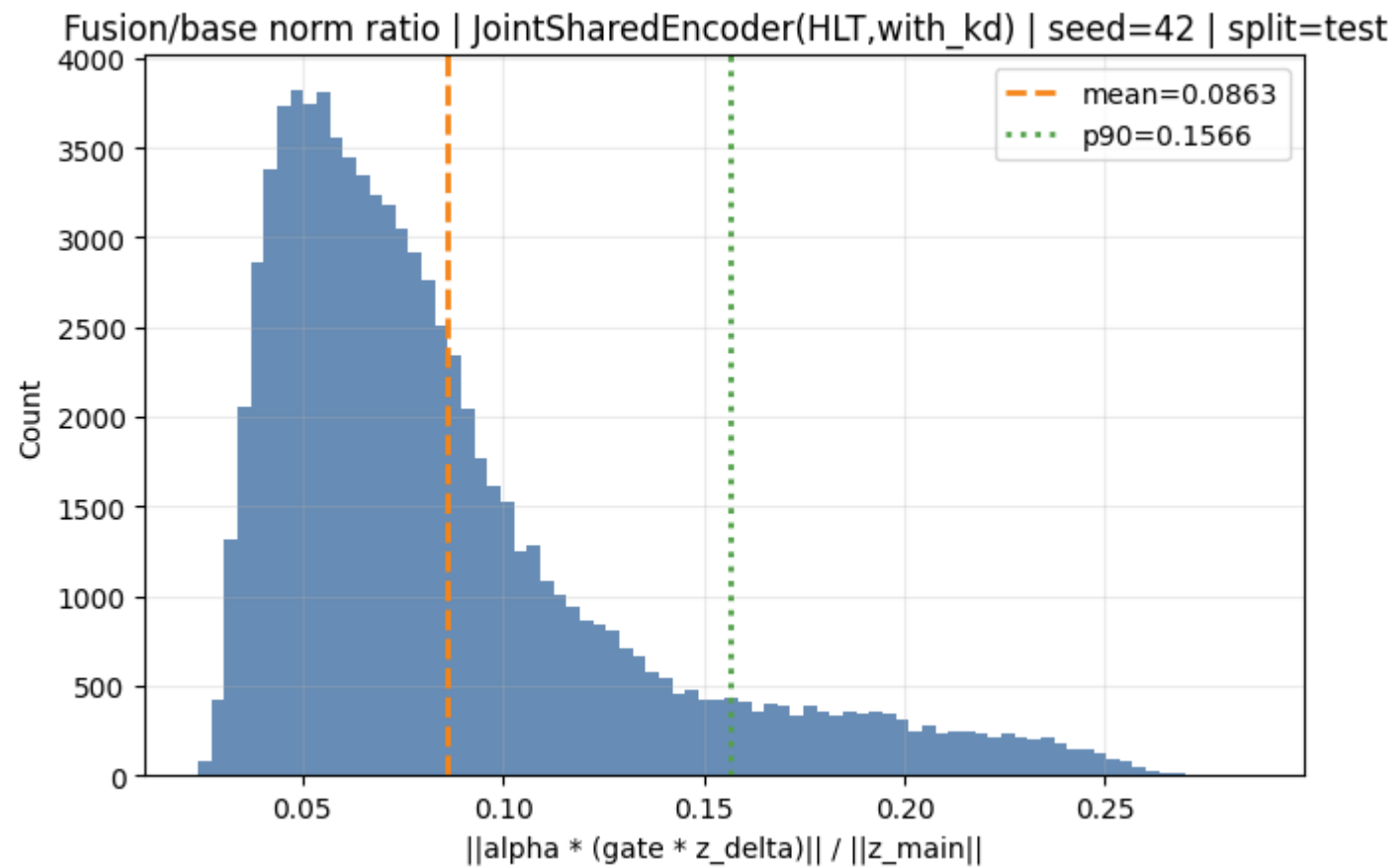
Still, almost no improvement is observed.



	Teacher	HLT Baseline	HLT/Joint+kd	Offline-HLT Gap Recovery
FPR @ 30%TPR	0.2181%	0.5781%	0.4774%/0.5005%	26.359%/20.319%
FPR @ 50%TPR	0.9710%	2.1771%	1.9160%/1.9635%	20.350%/16.435%

The reconstructed features are indeed closer to the offline features than the original HLT features, especially for angular features.





Fusion has a very limited contribution to the final classifier

Reconstruction improves in terms of MSE, but classification performance does not improve. This suggests that MSE-based reconstruction may not recover the information most relevant for the classification task. (Physics consistency loss does not lead to significant improvements and unordered set losses (e.g. Chamfer loss) largely reduce to MSE-type objectives in this setting.)

The key challenge becomes designing losses or architectures that reconstruct information relevant for classification, which first requires identifying what information the classifier actually relies on.

Although the controlled study of smearing effects shows that angular smearing has the largest impact on classification AUC, increasing the angular reconstruction loss weight still does not improve performance. This suggests that the key classification-relevant information may not be captured by simply emphasizing angular.

On the other hand, the influence of the reconstruction task on classification remains consistently weak. It is therefore unclear whether the tested reconstruction objectives fail to recover useful classification-relevant information, or whether reconstruction itself provides limited benefit to classification. However, the controlled study of smearing effects previously showed that reducing smearing does improve classification AUC.

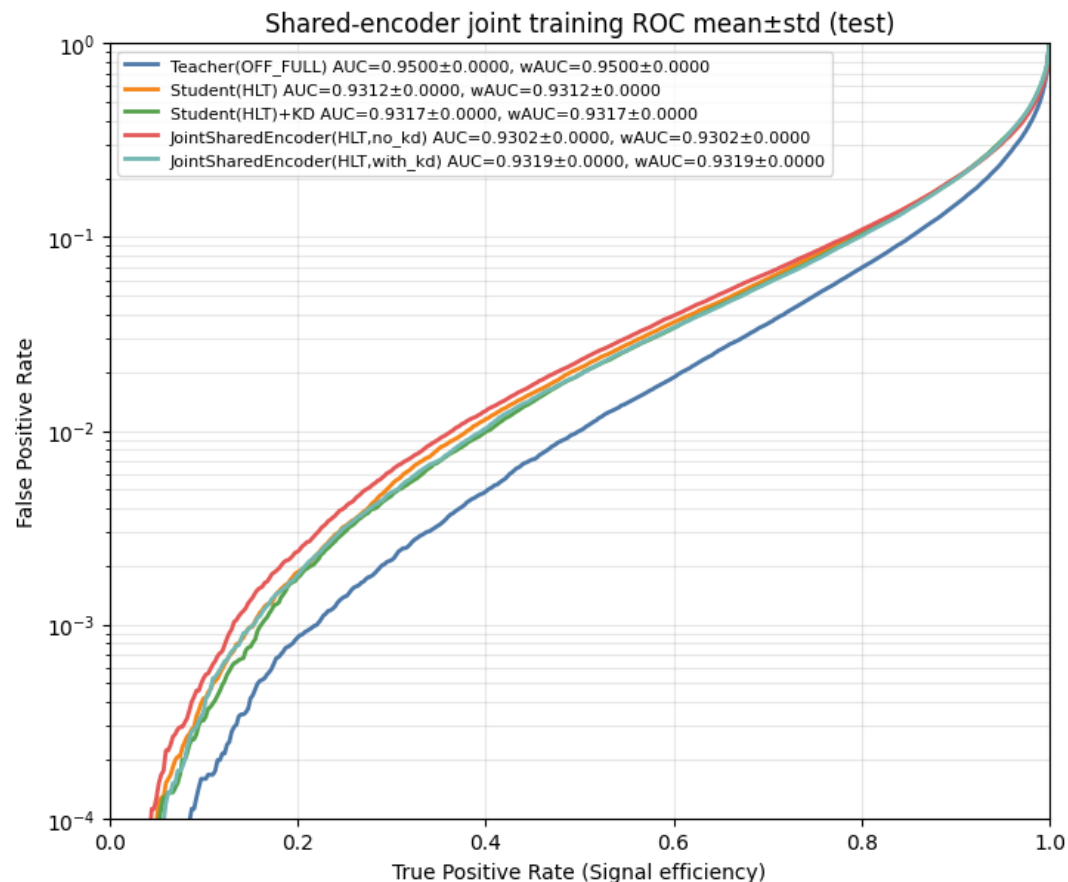
Additional Experiments

Multi-View Smearing per Epoch

1.5M (1M*3/0.25M/0.25M)

Each training jet is independently re-smearred three times per epoch, effectively tripling the training exposure to smear variations without changing the underlying train split.

No significant improvement is observed.



	Teacher	HLT Baseline	HLT/Joint+kd	Offline-HLT Gap Recovery
FPR @ 30%TPR	0.2150%	0.5268%	0.4621%/0.4804%	20.769%/14.872%
FPR @ 50%TPR	0.9977%	2.1041%	1.9466%/1.9439%	14.235%/14.480%

Teacher Embedding Distance Analysis

Teacher embedding distance refers to the distance, in the teacher embedding space, between the HLT/reconstructed inputs and the offline targets. In principle, smaller is better.

Although HLT has a larger embedding distance overall, it still achieves higher classification AUC. (compared to joint_no_kd)

Even in cases where HLT is correct and the joint model is wrong, HLT often has a larger distance.

This suggests that teacher embedding distance is not a reliable metric for reconstruction quality.

